

SPL Attack Detection Cheat Sheet

23 common attack techniques across the kill chain, with detection logic and example SPL searches.

Kill Chain Stage	Attack / Technique	MITRE ID	Detection Logic	SPL Example
Initial Access	Brute Force Authentication	T1110	Repeated failed logons against a single account in a short window.	index=main source="XmlWinEventLog:Security" EventID=4625 bin _time span=10m stats count as Fails by TargetAccount, _time where Fails>10
Initial Access	Password Spray	T1110.003	One source IP failing logon across many distinct accounts.	index=main source="XmlWinEventLog:Security" EventID=4625 bin _time span=10m stats dc(TargetAccount) as Accts by src_ip, _time where Accts>10
Execution	Suspicious PowerShell	T1059.001	Encoded, hidden-window, or download-cradle PowerShell commands.	index=main source="*Sysmon/Operational" EventID=1 Image="powershell.exe" (CommandLine="*-enc*" OR CommandLine="*-EncodedCommand*" OR CommandLine="*IEX*" OR CommandLine="*DownloadString*")
Execution	Suspicious Cmd Execution	T1059.003	cmd.exe spawning reconnaissance or unusual child processes.	index=main source="*Sysmon/Operational" EventID=1 ParentImage="cmd.exe" (Image="*whoami.exe" OR Image="*net.exe" OR Image="*nslookup.exe")
Execution	LOLBAS Abuse	T1218 / T1105	Signed native binaries invoked with an HTTP reference (download).	index=main source="*Sysmon/Operational" EventID=1 (Image="*certutil.exe" OR Image="*bitsadmin.exe" OR Image="*mshta.exe" OR Image="*regsvr32.exe") CommandLine="*http"
Execution	Ingress Tool Transfer	T1105	A download command resulted in a new file written to disk.	index=main source="*Sysmon/Operational" EventID=11 (TargetFilename="C:\Windows*.exe" OR TargetFilename="C:\Temp*.exe")
Persistence	Registry Run Key	T1547.001	New value written to a common autorun registry path.	index=main source="*Sysmon/Operational" EventID=13 TargetObject="*\CurrentVersion\Run"
Persistence	Scheduled Task Creation	T1053.005	New scheduled task created via schtasks.	index=main source="*Sysmon/Operational" EventID=1 Image="schtasks.exe" CommandLine="*/create"
Persistence	New Service Creation	T1543.003	New service installed, especially with a short/random name.	index=main source="XmlWinEventLog:System" EventCode=7045 regex ServiceName="^[A-Za-z]{4,8}\$"
Defense Evasion	Clearing Event Logs	T1070.001	wevtutil used to clear Security/System/Application logs.	index=main source="*Sysmon/Operational" EventID=1 Image="wevtutil.exe" CommandLine="* cl *"
Defense Evasion	Disabling Security Tools	T1562.001	PowerShell command disabling Defender real-time protection.	index=main source="*Sysmon/Operational" EventID=1 (CommandLine="*Set-MpPreference*" OR CommandLine="*DisableRealtimeMonitoring*")
Defense Evasion	Masquerading	T1036	Well-known process name running from an unexpected file path.	index=main source="*Sysmon/Operational" EventID=1 Image="svchost.exe" NOT Image="C:\Windows\System32*"

Kill Chain Stage	Attack / Technique	MITRE ID	Detection Logic	SPL Example
Credential Access	LSASS Credential Dumping	T1003.001	A process opens a handle to LSASS memory (e.g. Mimikatz-style tools).	index=main source="*Sysmon/Operational" EventID=10 TargetImage="*lsass.exe"
Credential Access	Kerberoasting	T1558.003	Kerberos service ticket requested using weak (RC4) encryption.	index=main source="XmlWinEventLog:Security" EventID=4769 TicketEncryptionType=0x17
Lateral Movement	PsExec-Style Service Creation	T1021.002	New service pointing to an executable with a randomized name.	index=main source="XmlWinEventLog:System" EventCode=7045 regex ImagePath=".*\\[A-Za-z]{8}\\\.exe\$"
Lateral Movement	Named Pipe Lateral Movement	T1021.002	Named pipes matching known remote execution tool naming.	index=main source="*Sysmon/Operational" (EventID=17 OR EventID=18) (PipeName="*RemCom*" OR PipeName="*PSEXESVC*" OR PipeName="*status*")
Lateral Movement	WMI Lateral Movement	T1047	wmic.exe used to execute commands against a remote host.	index=main source="*Sysmon/Operational" EventID=1 Image="*wmic.exe" CommandLine="*/node:*
Lateral Movement	Pass-the-Hash	T1550.002	NTLM network logon that skips normal interactive authentication.	index=main source="XmlWinEventLog:Security" EventID=4624 LogonType=9 AuthenticationPackageName=NTLM
Lateral Movement	RDP Lateral Movement	T1021.001	Interactive RDP logon (Type 10) to a host.	index=main source="XmlWinEventLog:Security" EventID=4624 LogonType=10
Command and Control	C2 Beaconsing	T1071	Host making periodic connections to the same external IP.	index=main source="*Sysmon/Operational" EventID=3 bin _time span=1h stats count as Conns by DestinationIp, _time where Conns>50
Command and Control	DNS Tunneling	T1071.004	Unusually high volume of DNS queries from a single host.	index=main source="*Sysmon/Operational" EventID=22 bin _time span=1h stats count as QueryCount by Computer, _time where QueryCount>200
Exfiltration	Large Outbound Transfer	T1041	Abnormally large outbound data volume from a single host.	index=main source="*Sysmon/Operational" EventID=3 bin _time span=1h stats sum(SentBytes) as Bytes by Computer, _time where Bytes>500000000
Domain / AD	DCSync Attack	T1003.006	Directory replication request from a non-domain-controller account.	index=main source="XmlWinEventLog:Security" EventID=4662 Properties="*1131f6aa-9c07-11d1-f79f-00c04fc2dcd2"

Note: Searches use generic index/source names (index=main, XmlWinEventLog:Security, Sysmon/Operational) matching common Splunk lab setups. Adjust index, source, and field names to match your own environment. Thresholds (e.g. Fails>10, Conns>50) are starting points, not fixed rules; tune them against your own environment's normal baseline.